

Odd End Semester Examination 2025

Name of the Program: **B.Tech. Biotechnology**

Semester: **5th**

Name of the Course: **Circular Economy and Sustainability**

Course Code: **TBT509c**

Time: **3hrs**

Maximum Marks: **100**

Note:

- Answer **all the questions** by choosing *any two of the sub questions*.
- Each question carries **20 marks**

Q1														
(a)	Match the following: (20 marks)	CO1												
	<table><tr><td>i) Take-Make-Waste Model</td><td>A.) Returning products for refurbishment in supply chains</td></tr><tr><td>ii) Circular Economy</td><td>B.) Converting waste into reusable polymers</td></tr><tr><td>Integrated Organic Farming</td><td>C.) Linear economic approach leading to resource depletion</td></tr><tr><td>iii) Systems</td><td>D.) Holistic nutrient cycling in agriculture</td></tr><tr><td>iv) Waste Recycling Tech</td><td>E.) Promotes resource efficiency and waste minimization</td></tr><tr><td>v) Reverse Logistics</td><td></td></tr></table>		i) Take-Make-Waste Model	A.) Returning products for refurbishment in supply chains	ii) Circular Economy	B.) Converting waste into reusable polymers	Integrated Organic Farming	C.) Linear economic approach leading to resource depletion	iii) Systems	D.) Holistic nutrient cycling in agriculture	iv) Waste Recycling Tech	E.) Promotes resource efficiency and waste minimization	v) Reverse Logistics	
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OR														
(b)	Explain the core principles of the circular economy and contrast them with the linear "take-make-waste" model. Illustrate how these principles address sustainability challenges using a real-world example from resource depletion.													
OR														
(c)	Analyze the limitations of the linear economic model in terms of resource efficiency and waste generation. Discuss key economic and environmental impacts.													
Q2														
(a)	Fill in the blanks: (20 marks)	CO2												
	i) Life Cycle Assessment (LCA) measures a product's impact from _____ to grave. ii) Closed-loop supply chains incorporate _____ logistics to recover used products efficiently. iii) Waste management technologies include advanced _____ processes to convert plastics into fuels. iv) _____ is the world's first 100% organic farming state. v) Circular food processing emphasizes zero-waste techniques like converting waste into _____.													
OR														
(b)	Apply the concept of closed-loop supply chains and reverse logistics to design a practical system for a consumer electronics company.													
OR														
(c)	Describe various business models for the circular economy, such as leasing, sharing, and remanufacturing.													
Q3														
(a)	Apply digital technologies like IoT and big data to optimize a circular supply chain in industry. (20 marks)	CO3												
OR														
(b)	Apply stakeholder engagement strategies to collaborate on a circular initiative.													
OR														

(c)	Explain planned obsolescence as a deliberate design strategy and its implications for resource depletion.	
Q4	(20 marks)	
(a)	Give brief answers to these questions: i) Define the circular economy with schematic diagram. ii) What is reverse logistics?	CO4
	OR	
(b)	Apply precision farming and digital technologies to design a circular model for crop production on a small farm.	
	OR	
(c)	Describe how biological nutrients support regenerative cycles in the biosphere, contrasting them with the industrial recycling loops of technical nutrients.	
Q5	(20 marks)	
(a)	What is Integrated Organic Farming Systems (IOFS)? How can agricultural waste generate revenue streams?	CO5
	OR	
(b)	Apply LCA principles to assess the circularity of a product like a smartphone. Break down the assessment into cradle-to-grave phases and recommend improvements.	
	OR	
(c)	State whether each statement is True or False: i) Product design for circularity includes strategies like design for disassembly to facilitate reuse. ii) Blockchain technology in food systems enhances traceability from farm to consumer. iii) Compostable packaging in agriculture increases landfill contributions from food waste. iv) Reverse logistics only involves forward supply chains without recovery mechanisms. v) Stakeholder engagement is unnecessary for successful circular economy collaborations.	